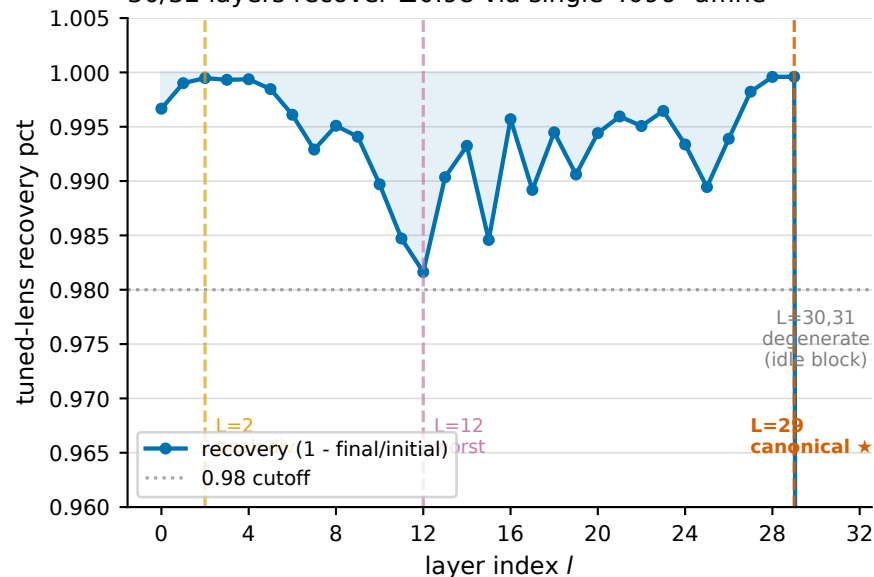


(a)**Where does the model
LOOK?**

attention maps

**What does the model
ENCODE?**

sparse autoencoders

**What does the model
PREDICT?**embedding distance,
sequence likelihood**Where does the model
THINK DEEPLY? ★**gDTR (this work)
layer-wise settling depth**(b)****NLP-DTR (Chen et al. 2026)**JSD lens, $\gamma=0.5$, $|V|\approx 100k$
GPT-OSS, DeepSeek-R1, Qwen3**gDTR (this work)**JSD + cosine UR lens
q70 calibration $\gamma_{\text{cos}}=0.40$
Evo 2 / HyenaDNA / NT / DNABERT-2**Three adaptation challenges:**C1 small vocabulary $|V| \leq 512 \rightarrow$ q70 calibrationC2 hybrid architectures $\rightarrow$ block-type-aware lensC3 untied $\text{lm_head} + \text{idle L31} \rightarrow$ tuned lens, tap = L29*Four axes of mechanistic interpretability for genomic CLMs***(c)**30/32 layers recover ≥ 0.98 via single 4096^2 affine**(d)**(d) Calibration: raw lens fails $\rightarrow$ tuned + running-min